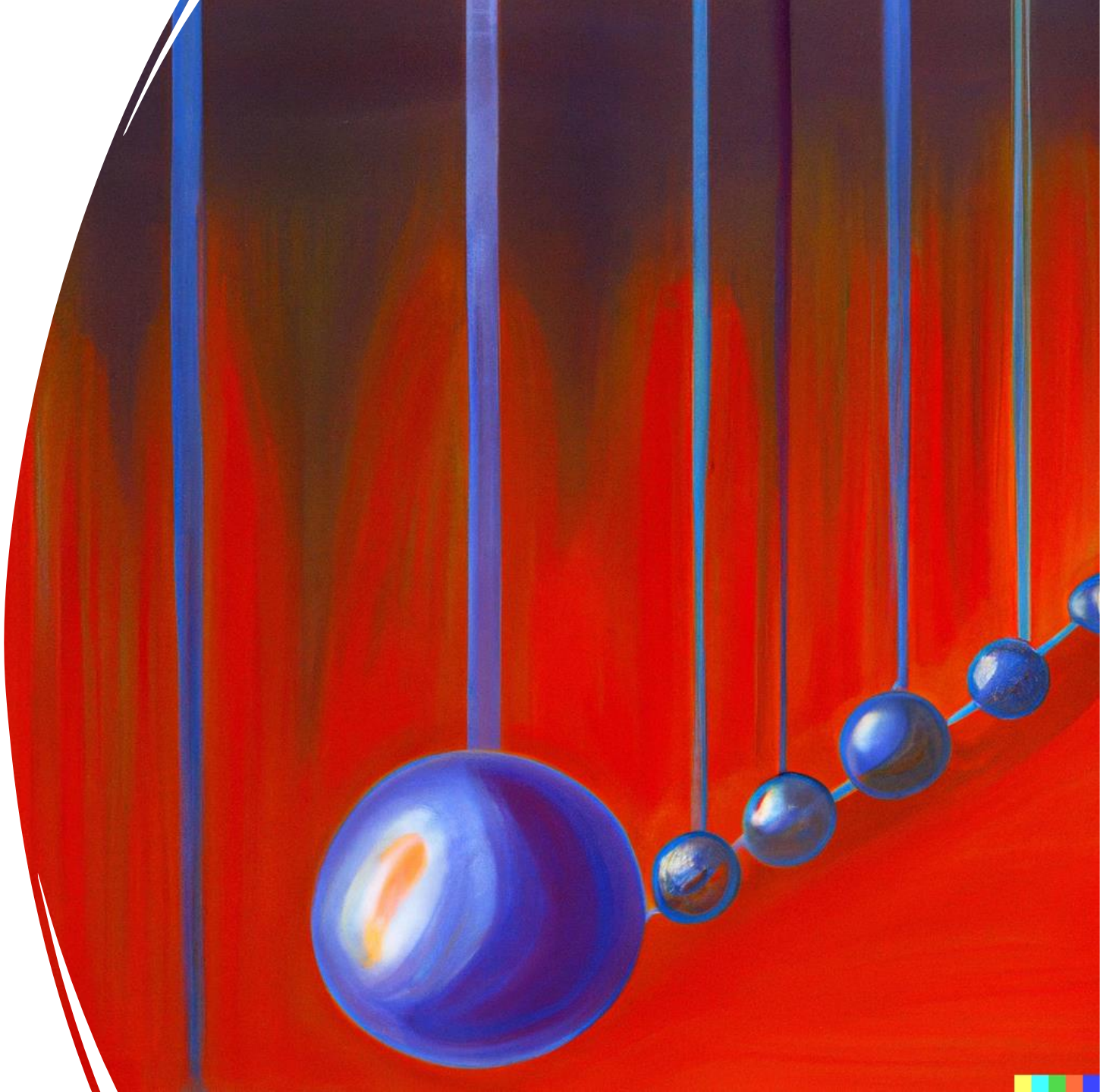
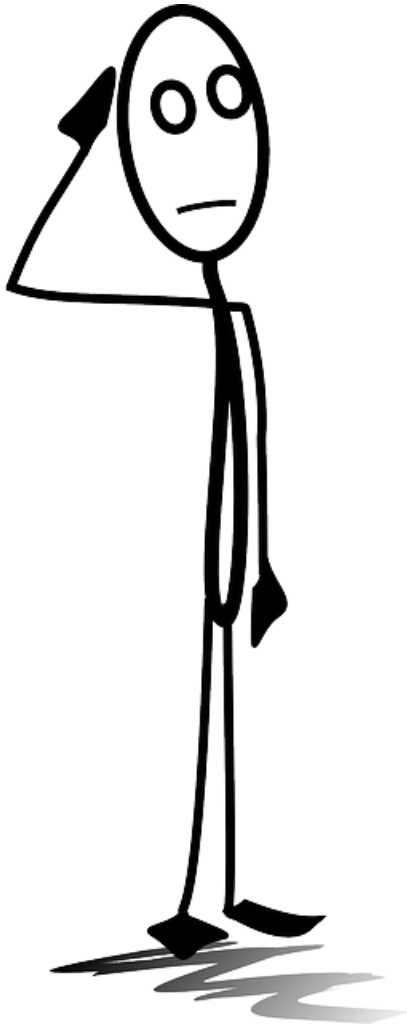
PHYSICS I
WEEK 3

Siddharth Dhomkar
Department of Physics
IIT Madras
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Office: HSB 116B



WEEK 3: OUTLINE

- Review:
  - Conservation of angular momentum: Taylor 3.5
  - Kinetic Energy: Taylor 4.1
  - Potential Energy: Taylor 4.3
- Simple harmonic motion: Taylor 5.1, 5.2



REVIEW:

Newton's Second Law (Inertial frame)

$$\boldsymbol{F} = m\boldsymbol{a} \Rightarrow \boldsymbol{F} = m\ddot{\boldsymbol{r}} \Rightarrow \boldsymbol{F} = \dot{\boldsymbol{p}}$$

Cartesian

$$F_x = m\ddot{x}$$
$$F_y = m\ddot{y}$$
$$F_z = m\ddot{z}.$$

$$\dot{x}(t) = \int \ddot{x}(t)\, dt = v_0 + \frac{F_0}{m} t$$

$$x(t) = \int \dot{x}(t) dt = x_0 + v_0 t + \frac{F_0}{2m} t^2$$

x_0, v_0 *Initial Conditions*

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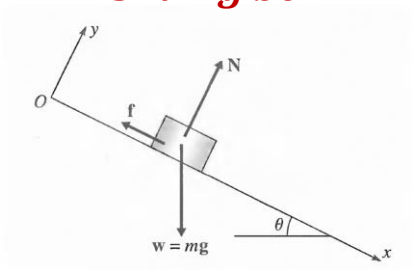
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x_0, v_0 *Initial Conditions*

Sliding box



$$\ddot{x} = g(\sin\theta - \mu\cos\theta)$$

$$x(t) = \tfrac{1}{2} g(\sin\theta - \mu\cos\theta)t^2$$

REVIEW:

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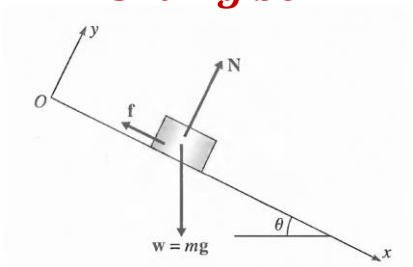
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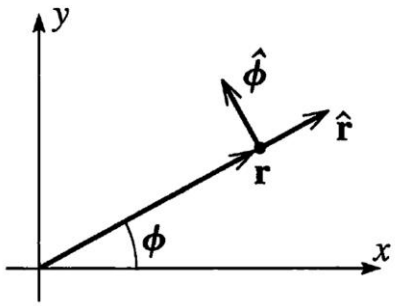
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Polar

$$F_r = m(\ddot{r} - r\dot{\phi}^2)$$
$$F_\phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi})$$

$$x = r\cos\phi$$
$$y = r\sin\phi$$

$$r = \sqrt{x^2 + y^2}$$
$$\phi = \arctan(y/x)$$



$$\mathbf{a} = \left(\ddot{r} - r\dot{\phi}^2\right)\hat{\mathbf{r}} + \left(r\ddot{\phi} + 2\dot{r}\dot{\phi}\right)\hat{\boldsymbol{\phi}}$$

REVIEW:

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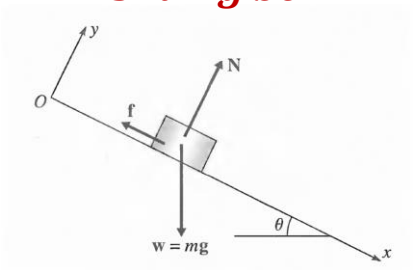
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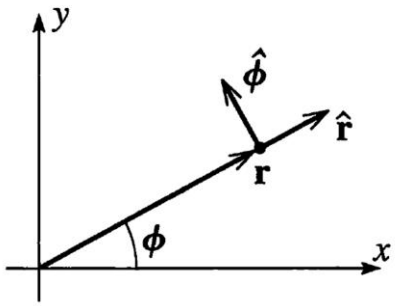
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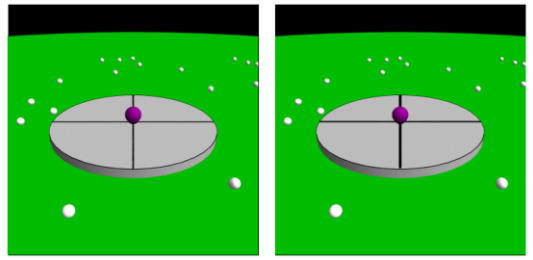


$$\mathbf{a} = \left(\ddot{r} - r\dot{\phi}^2\right)\hat{\mathbf{r}} + \left(r\ddot{\phi} + 2\dot{r}\dot{\phi}\right)\hat{\boldsymbol{\phi}}$$

Newton's Second Law (Non-inertial frame)

$$m\ddot{\boldsymbol{r}} = \boldsymbol{F} - m\boldsymbol{A}$$

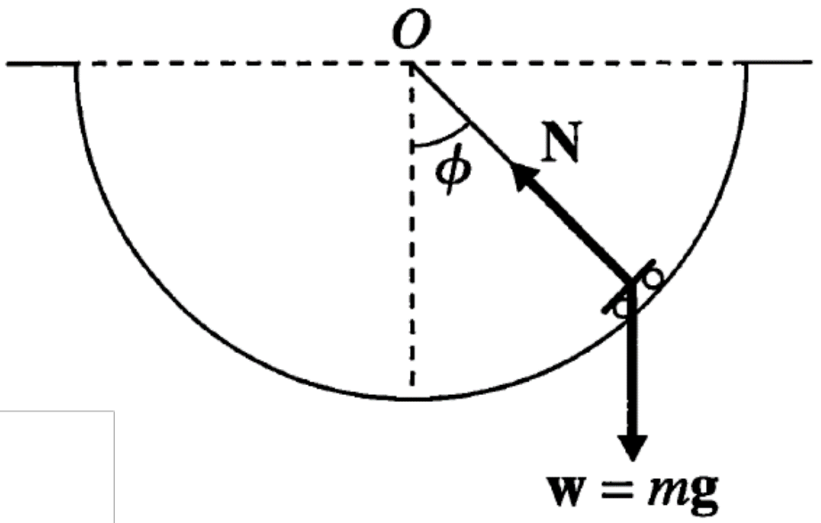
Inertial Force
(pseudo force)



NEWTON'S LAWS: POLAR COORDINATES

EXAMPLE 1.2 An Oscillating Skateboard

A "half-pipe" at a skateboard park consists of a concrete trough with a semicircular cross section of radius $R = 5$ m, as shown in Figure 1.14. I hold a frictionless skateboard on the side of the trough pointing down toward the bottom and release it. Discuss the subsequent motion using Newton's second law. In particular, if I release the board just a short way from the bottom, how long will it take to come back to the point of release?



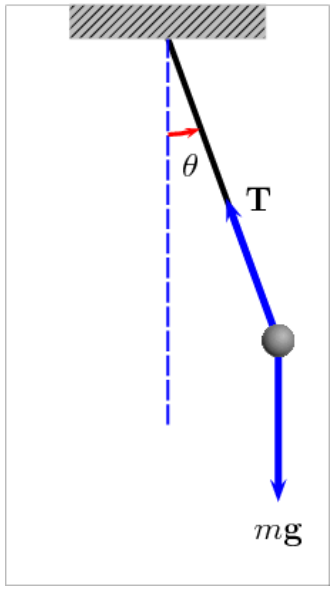
$$\ddot{\phi} = -\frac{g}{R}\sin\phi.$$

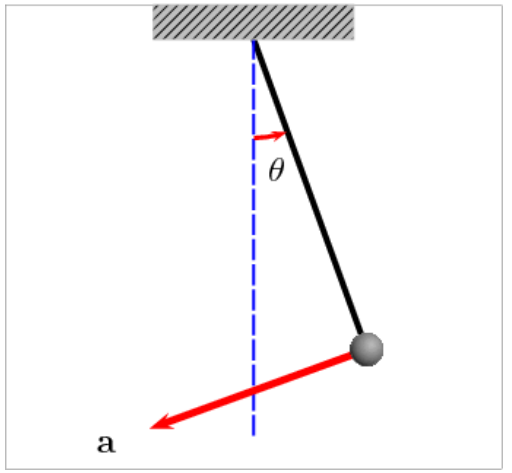
Small angle approximation

$$\omega = \sqrt{\frac{g}{R}}$$

Equation of motion

$$\ddot{\phi} = -\omega^2\phi$$





Solution

$$\phi(t) = A\sin(\omega t) + B\cos(\omega t)$$

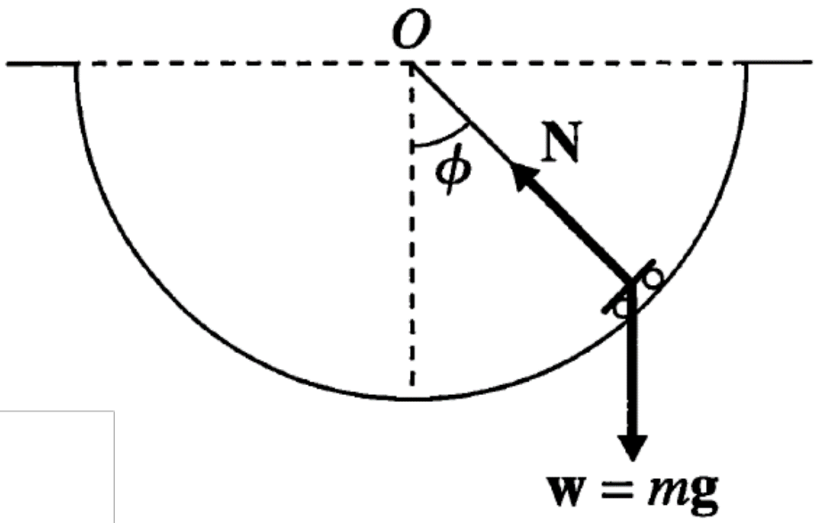
Initial Conditions

$$\phi_0(t) = 0 \quad \dot{\phi}_0(t) = 0$$

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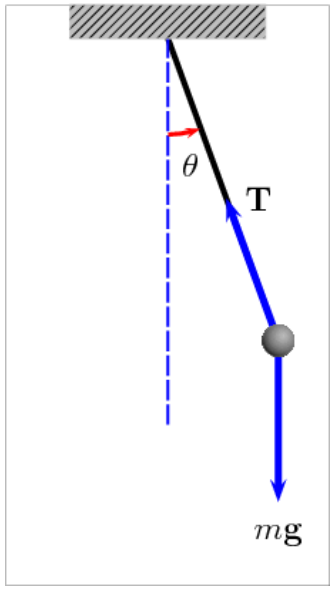
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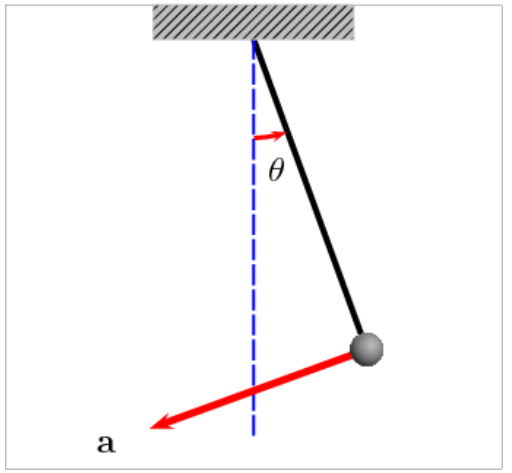
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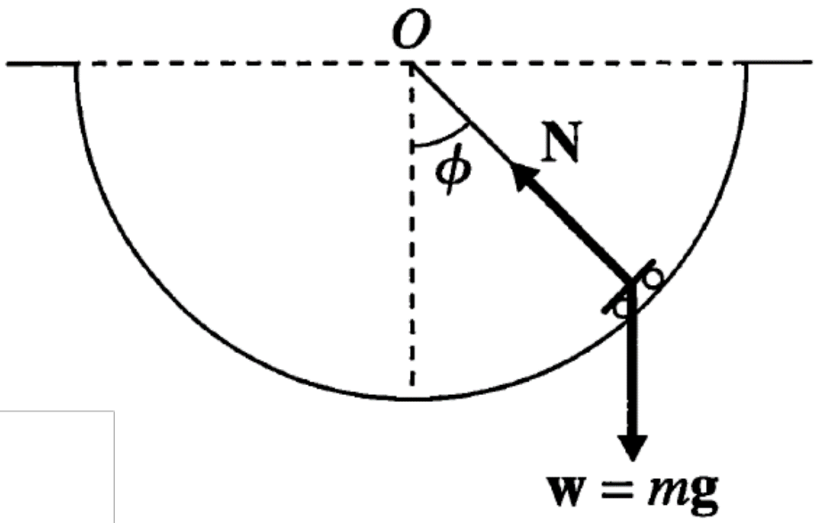
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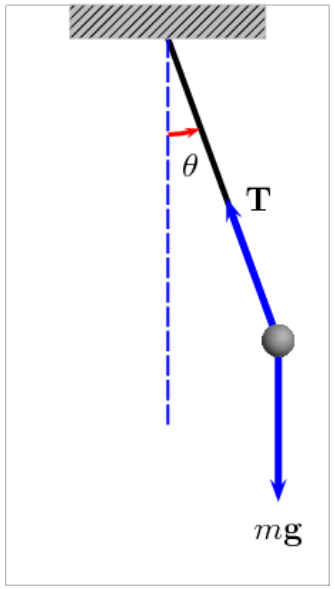
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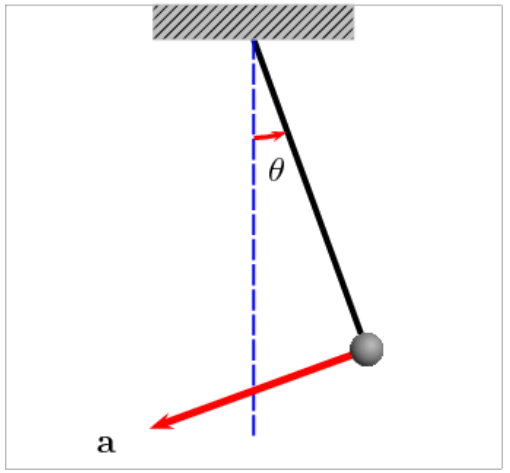
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$$\tau = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R}{g}}$$

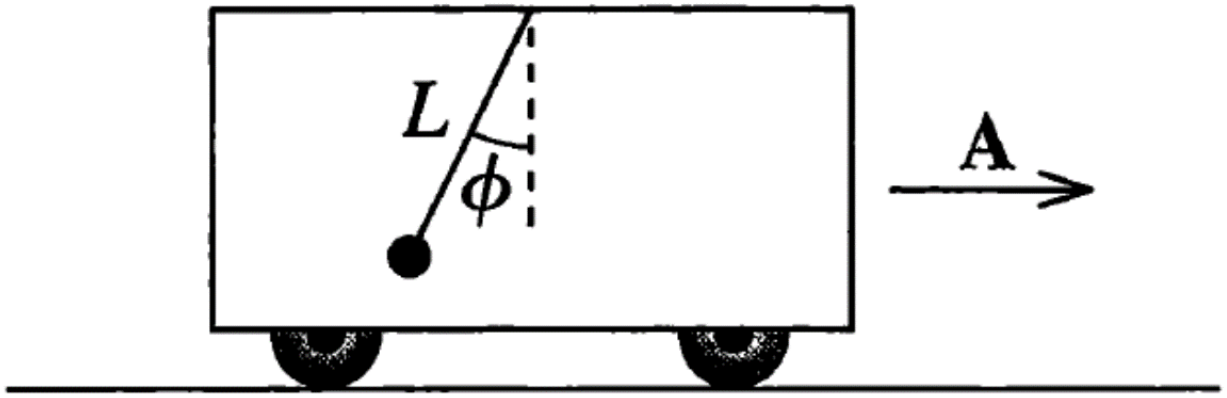
NEWTON'S LAWS: NON-INERTIAL FRAMES

EXAMPLE 9.1 A Pendulum in an Accelerating Car

Consider a simple pendulum (mass m and length L) mounted inside a railroad car that is accelerating to the right with constant acceleration **A** as shown in Figure 9.1. Find the angle ϕ_{eq} at which the pendulum will remain at rest relative to the accelerating car and find the frequency of small oscillations about this equilibrium angle.

$$m\ddot{\boldsymbol{r}} = \boldsymbol{F} - m\boldsymbol{A}$$

Inertial Force
(pseudo force)



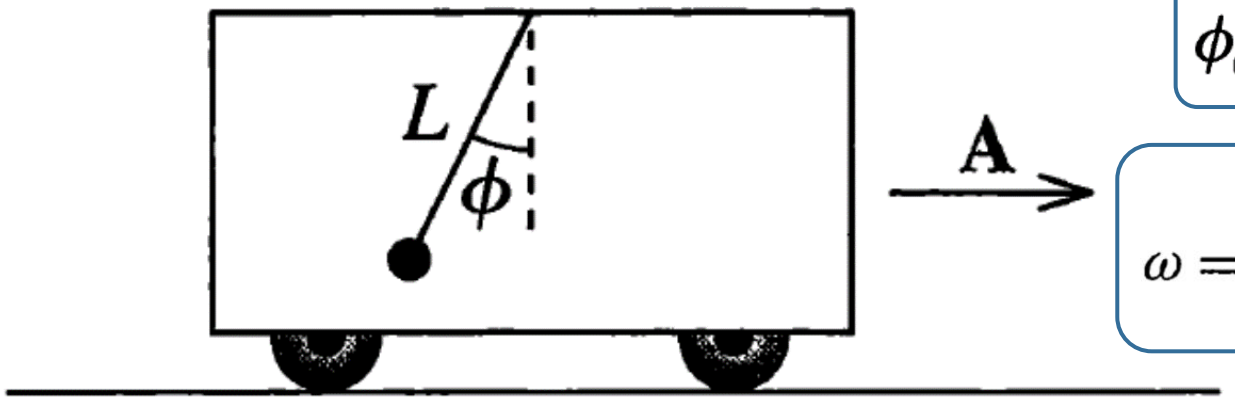
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Inertial Force
(pseudo force)



$$\phi_{eq} = \arctan(A/g)$$

$$\omega = \sqrt{\frac{g_{eff}}{L}} = \sqrt{\frac{\sqrt{g^2 + A^2}}{L}}$$

NEWTON'S LAWS: POLAR COORDINATES

Circular pendulum

Consider a circular pendulum of mass m and length L moving in the x-y-plane at an angle α with respect to the (vertical) z-axis.

(a) Find the angular frequency of the motion.

(b) For small α express the x-component of the force F_x in terms of its x-coordinate.

NEWTON'S LAWS: POLAR COORDINATES

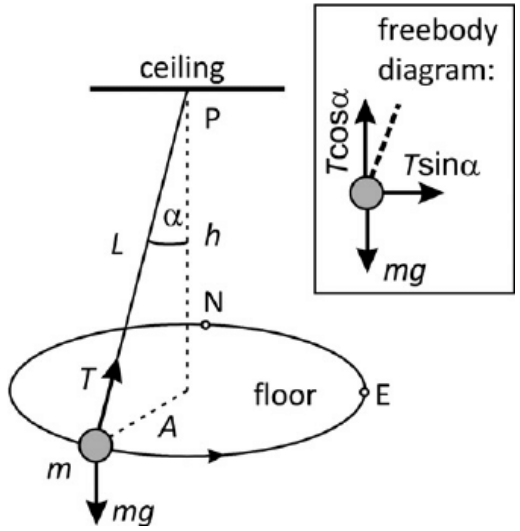
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NEWTON'S LAWS: POLAR COORDINATES

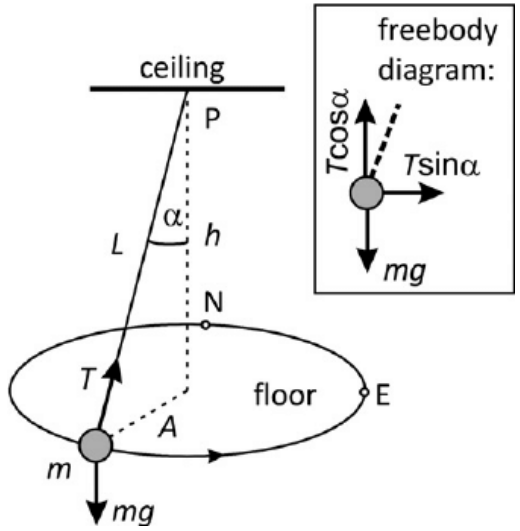
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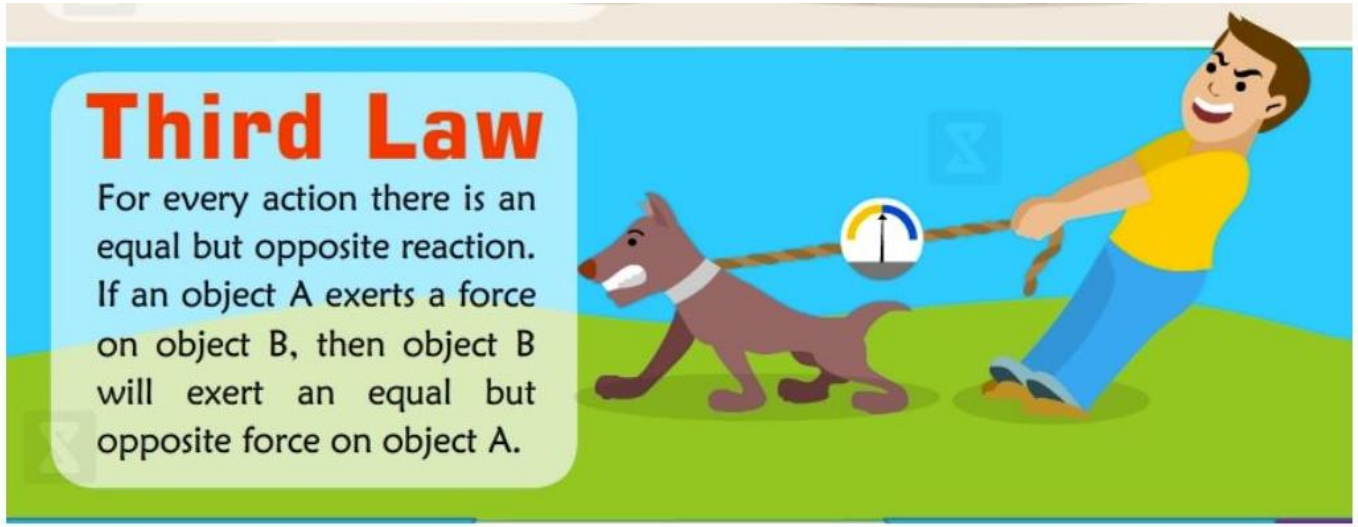
$$\mathbf{a} = \left(\ddot{r} - r\dot{\phi}^2\right)\hat{\mathbf{r}} + \left(r\ddot{\phi} + 2\dot{r}\dot{\phi}\right)\hat{\boldsymbol{\phi}}$$



$$\omega = \sqrt{\frac{g}{Lcos\alpha}}$$

$$F_x = -\mathrm{mg}\frac{x}{L}$$

NEWTON'S THIRD LAW



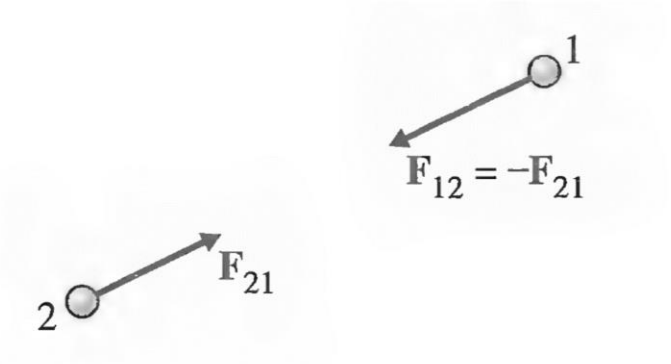


Figure 1.5 Newton's third law asserts that the reaction force exerted on object 1 by object 2 is equal and opposite to the force exerted on 2 by 1, that is, $\mathbf{F}_{12} = -\mathbf{F}_{21}$.

NEWTON'S THIRD LAW

Conservation of Momentum

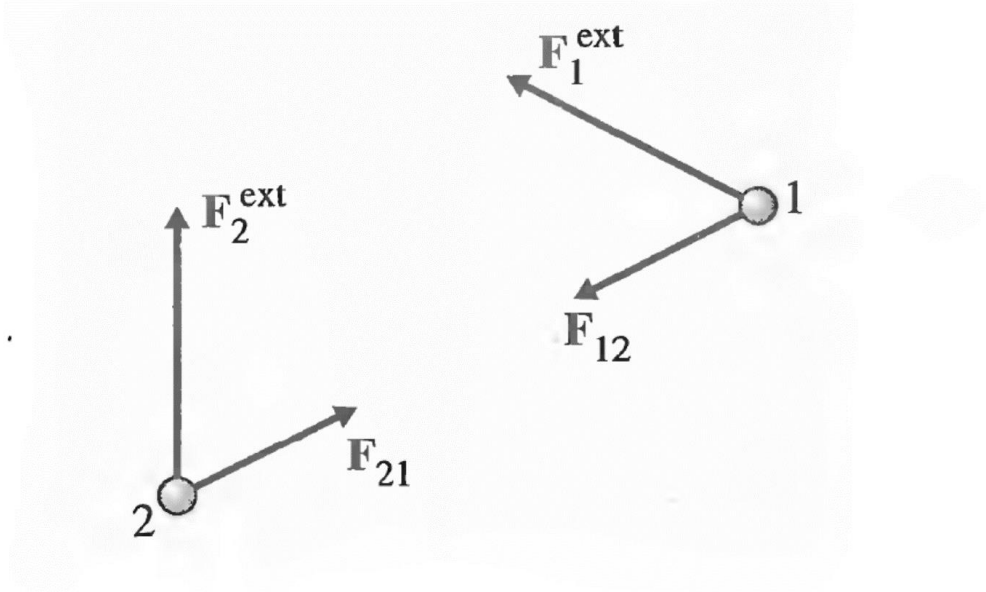


Figure 1.6 Two objects exert forces on each other and may also be subject to additional "external" forces from other objects not shown.

NEWTON'S THIRD LAW

Conservation of Momentum

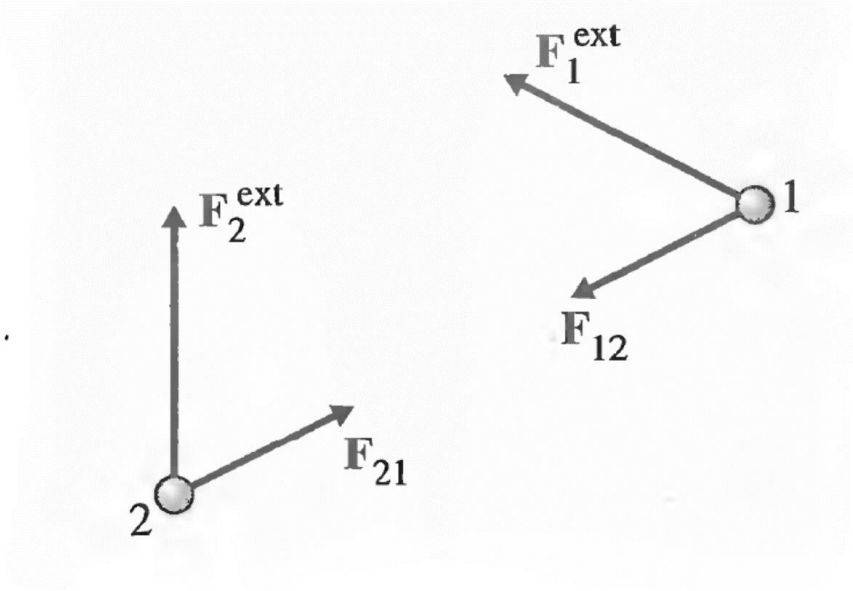


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$$\dot{\mathbf{P}} = \mathbf{F}_1^{\text{ext}} + \mathbf{F}_2^{\text{ext}} \equiv \mathbf{F}^{\text{ext}}$$

If $\mathbf{F}^{\text{ext}} = 0$, then $\mathbf{P} = \text{const.}$

NEWTON'S THIRD LAW

Conservation of Momentum

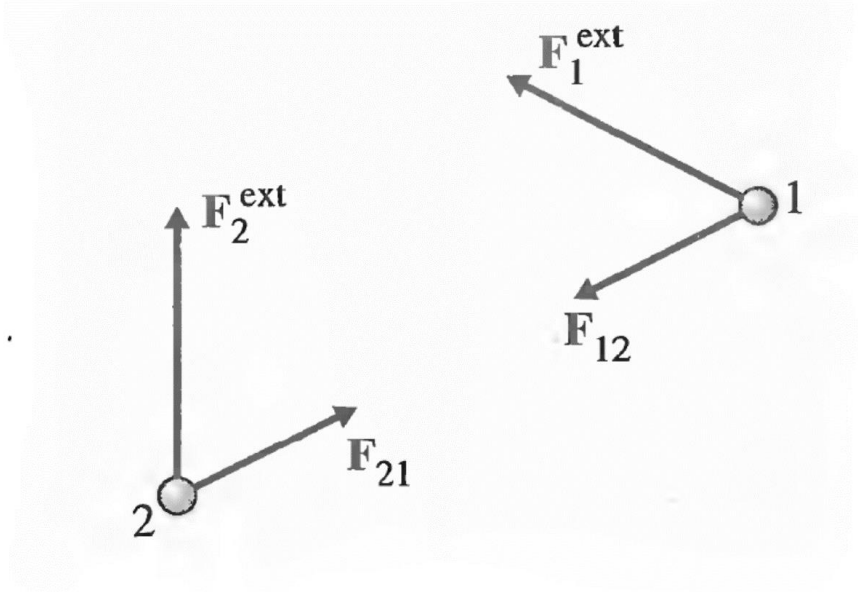


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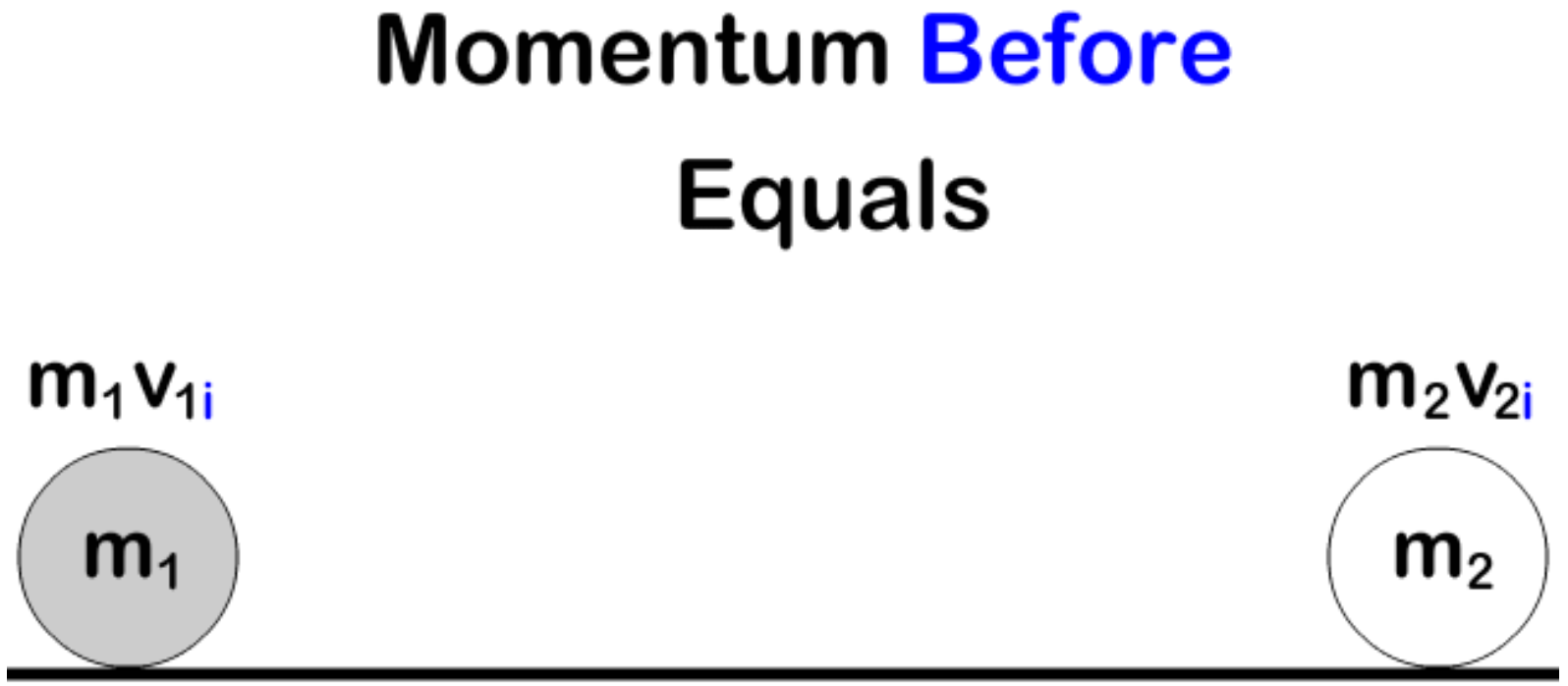
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For multi-particle system

$$\dot{\mathbf{P}} = \sum_{\alpha} \mathbf{F}_{\alpha}^{\text{ext}} \equiv \mathbf{F}^{\text{ext}}$$

NEWTON'S THIRD LAW

Conservation of Momentum



REVIEW: ENERGY

Kinetic Energy

$$T = \frac{1}{2}mv^2 \Rightarrow \frac{dT}{dt} = \mathbf{F}\cdot\mathbf{v} \Rightarrow dT = \mathbf{F}\cdot d\mathbf{r}$$

$$\Delta T \equiv T_2 - T_1 = \int_1^2 \mathbf{F}\cdot d\mathbf{r} \equiv W(1\rightarrow 2)$$

Work done by the force moving between point 1 and 2

REVIEW: ENERGY

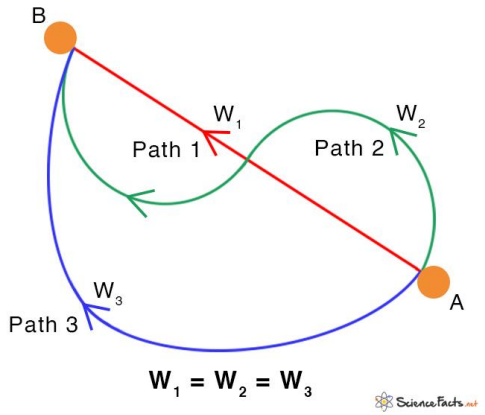
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Work done by the force moving between point 1 and 2

Conservative Force



REVIEW: ENERGY

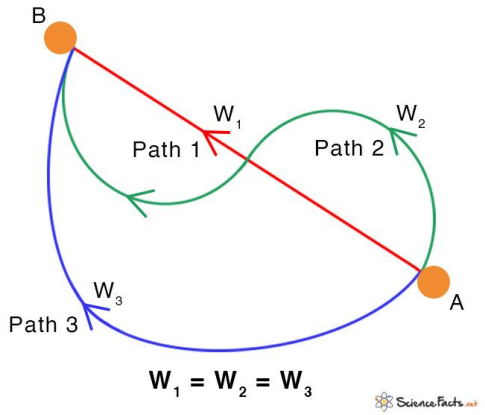
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Work done by the force moving between point 1 and 2

Conservative Force



Potential Energy

$$U(\mathbf{r}) = -W(\mathbf{r}_o \rightarrow \mathbf{r}) \equiv -\int_{\mathbf{r}_o}^{\mathbf{r}} \mathbf{F}(\mathbf{r}')\cdot d\mathbf{r}'$$

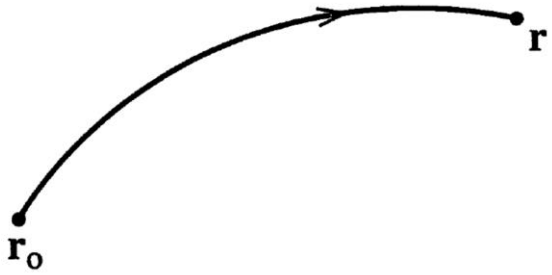
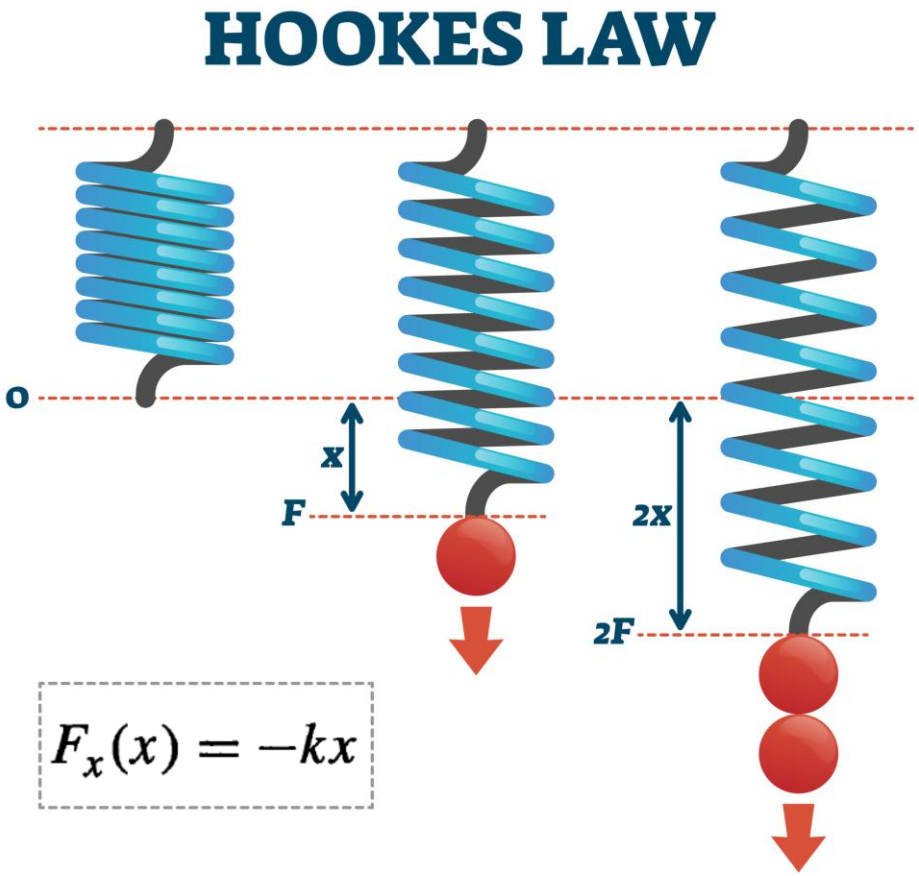


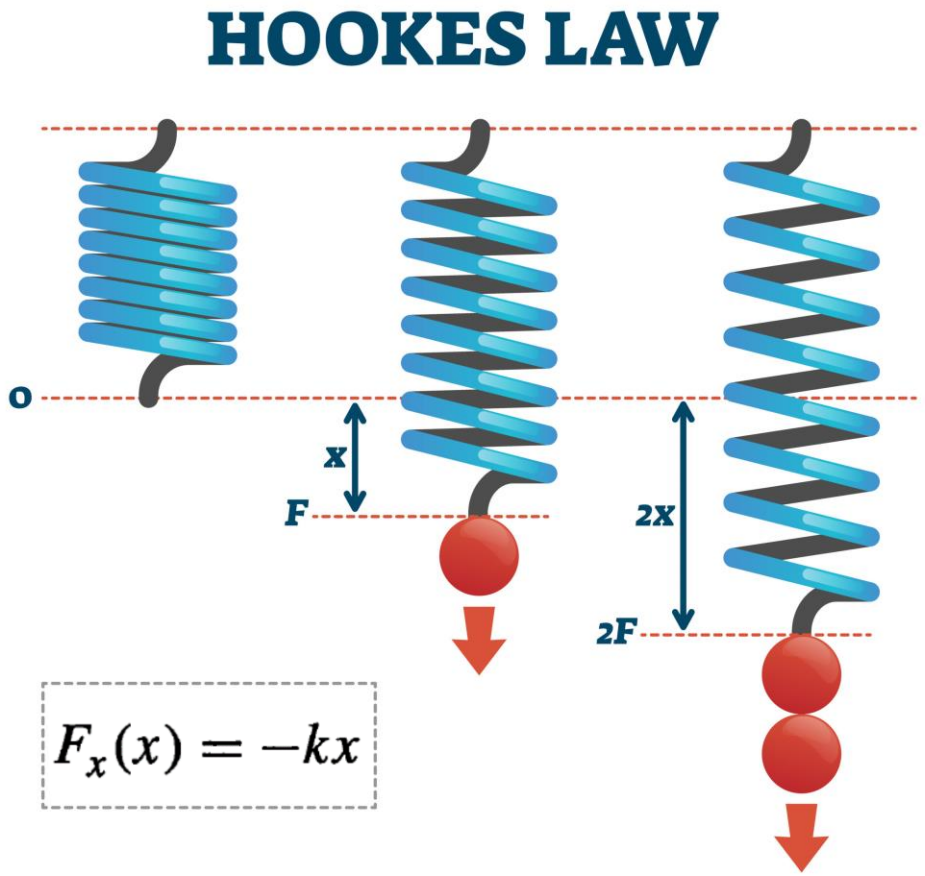
Figure 4.4 The potential energy $U(\mathbf{r})$ at any point $\mathbf{r}$ is defined as minus the work done by $\mathbf{F}$ if the particle moves from the reference point $\mathbf{r}_o$ to $\mathbf{r}$. This gives a well-defined function $U(\mathbf{r})$ only if this work is independent of the path followed — that is, the force is conservative.

$$E = \mathrm{KE} + \mathrm{PE} = T + U(\mathbf{r}) \qquad \Delta T = -\Delta U$$

SIMPLE HARMONIC MOTION



SIMPLE HARMONIC MOTION



$$F_x(x) = -kx$$

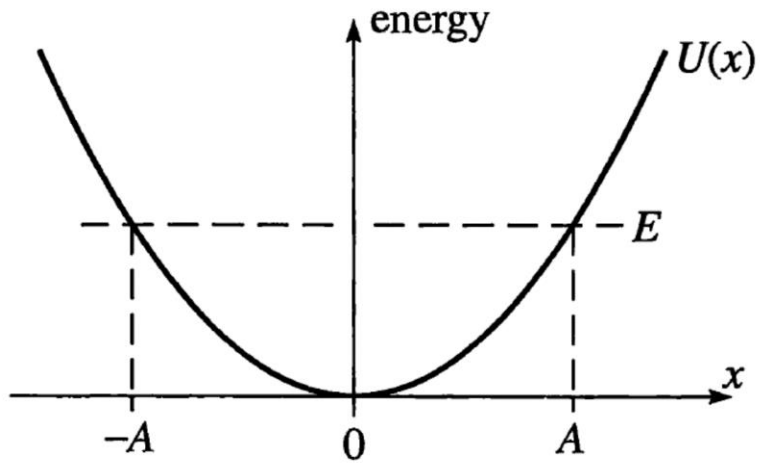
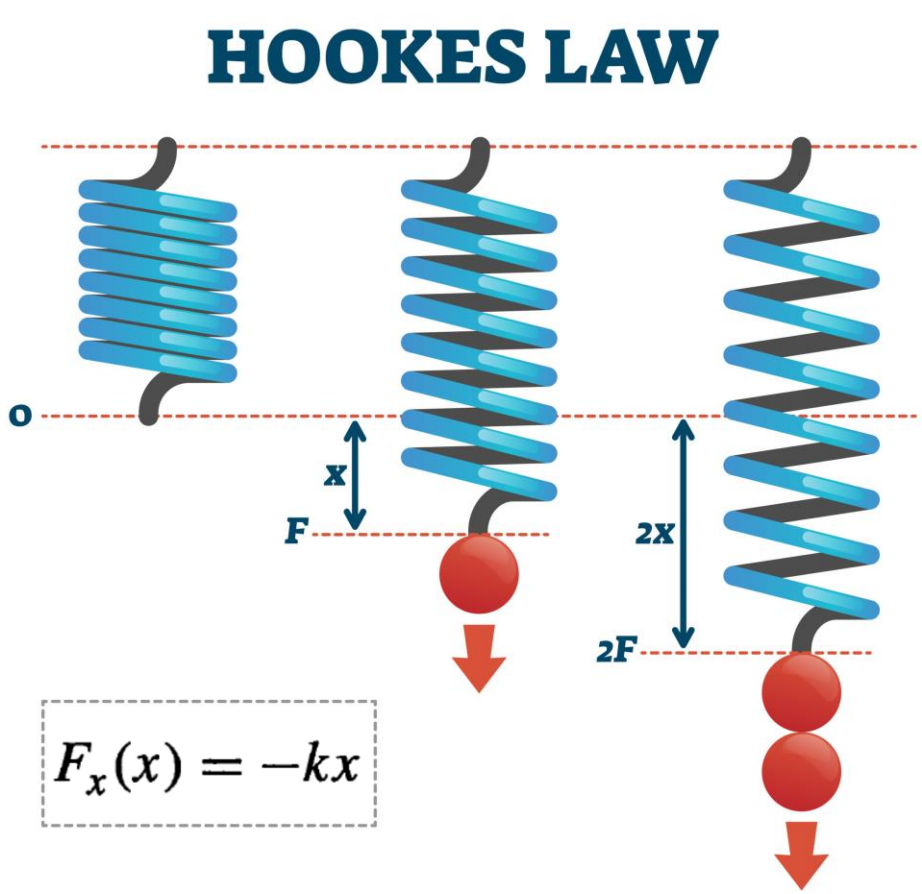


Figure 5.1 A mass m with potential energy $U(x) = \frac{1}{2}kx^2$ and total energy E oscillates between the two turning points at $x = \pm A$, where $U(x) = E$ and the kinetic energy is zero.

SIMPLE HARMONIC MOTION



$$F_x(x) = -kx$$

$$F(x) = -\frac{dU}{dx}$$

$$U(\mathbf{r}) = -W(\mathbf{r}_o \to \mathbf{r}) \equiv -\int_{\mathbf{r}_o}^{\mathbf{r}} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}'$$

$$U(x) = \tfrac{1}{2}kx^2$$

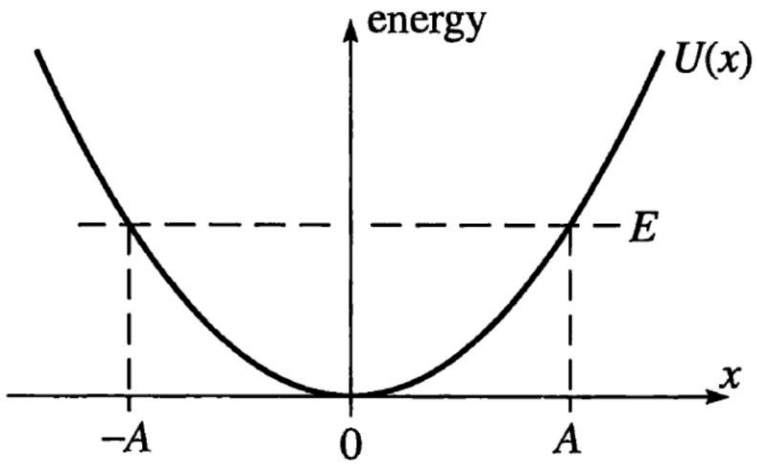


Figure 5.1 A mass m with potential energy $U(x) = \frac{1}{2}kx^2$ and total energy E oscillates between the two turning points at $x = \pm A$, where $U(x) = E$ and the kinetic energy is zero.

SIMPLE HARMONIC MOTION: SOLUTIONS

$$F_x(x) = -kx$$



$$m\ddot{x} = F_x = -kx$$

SIMPLE HARMONIC MOTION: SOLUTIONS

A second-order, linear, homogeneous differential equation

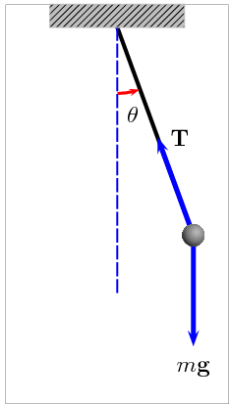
$$F_x(x) = -kx$$

$$\Downarrow$$

$$m\ddot{x} = F_x = -kx$$

$$\Downarrow$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$



$$\omega = \sqrt{\frac{k}{m}}$$

Equation of motion

$$\ddot{\phi} = -\omega^2 \phi$$

SIMPLE HARMONIC MOTION: SOLUTIONS

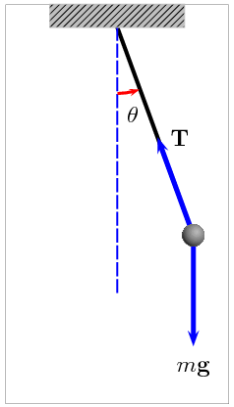
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A second-order, linear, homogeneous differential equation

General Solution

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

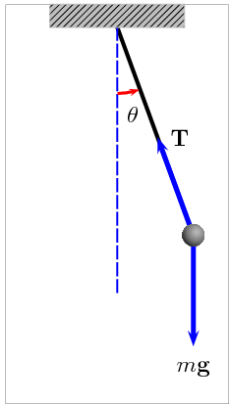
C_1 and C_2 depend on the initial conditions
(need to be chosen carefully to have real valued $x(t)$)

SIMPLE HARMONIC MOTION: SOLUTIONS

$$F_x(x) = -kx$$

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A second-order, linear, homogeneous differential equation

General Solution

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

C_1 and C_2 depend on the initial conditions (need to be chosen carefully to have real valued $x(t)$)

$$e^{\pm i\omega t} = \cos(\omega t) \pm i \sin(\omega t)$$

Euler's formula

$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$

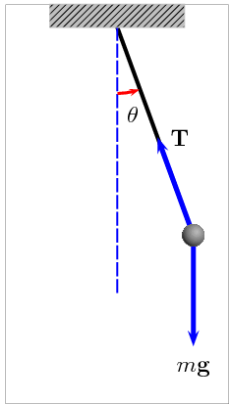
$B_1 = C_1 + C_2$ and $B_2 = i(C_1 - C_2)$

SIMPLE HARMONIC MOTION: SOLUTIONS

$$F_x(x) = -kx$$

$$m\ddot{x} = F_x = -kx$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$



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A second-order, linear, homogeneous differential equation

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Euler's formula

$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t) \qquad B_1 = C_1 + C_2 \quad \text{and} \quad B_2 = i(C_1 - C_2)$$

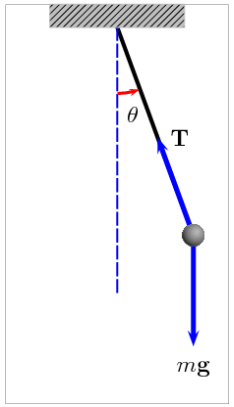
$$x(t) = x_o \cos(\omega t)$$ **Initial conditions $x = x_0$ & $v = 0$**

SIMPLE HARMONIC MOTION: SOLUTIONS

$$F_x(x) = -kx$$

$$m\ddot{x} = F_x = -kx$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$



$$\omega = \sqrt{\frac{k}{m}}$$

Equation of motion

$$\ddot{\phi} = -\omega^2 \phi$$

A second-order, linear, homogeneous differential equation

General Solution

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

C_1 and C_2 depend on the initial conditions (need to be chosen carefully to have real valued $x(t)$)

$$e^{\pm i\omega t} = \cos(\omega t) \pm i \sin(\omega t)$$

Euler's formula

$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t) \qquad B_1 = C_1 + C_2 \quad \text{and} \quad B_2 = i(C_1 - C_2)$$

$$x(t) = x_0 \cos(\omega t)$$ **Initial conditions $x = x_0$ & $v = 0$**

$$x(t) = \frac{v_0}{\omega} \sin(\omega t)$$ **Initial conditions $x = 0$ & $v = v_0$**

Period

$$\tau = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$